

LAB EXPERIMENT RECORD

Set 12: Website Traffic

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Date	06/10/2026
Day	Wednesday

AIM

To analyze website traffic data using **R and Tableau** by creating a line chart for daily page views, a bar chart for the top days based on click-through rate, a data table, and an interactive dashboard.

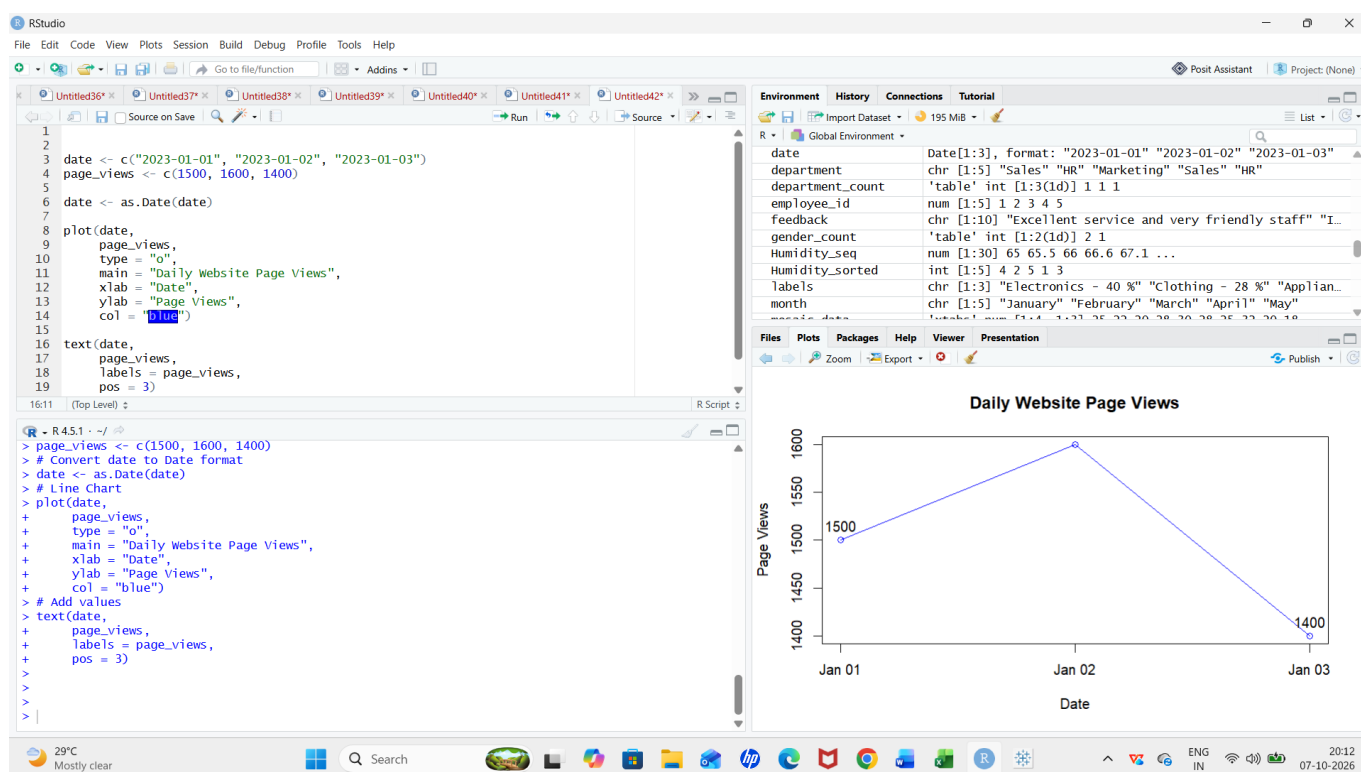
1. Create a line chart to visualize the trend in daily page views over time.

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
page_views <- c(1500, 1600, 1400)
```

```
date <- as.Date(date)
```

```
plot(date,
      page_views,
      type = "o",
      main = "Daily Website Page Views",
      xlab = "Date",
      ylab = "Page Views",
      col = "blue")
```

```
text(date,
      page_views,
      labels = page_views,
      pos = 3)
```



2. Generate a bar chart showing the top N days with the highest click-through rates.

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
ctr <- c(2.3, 2.7, 2.0)

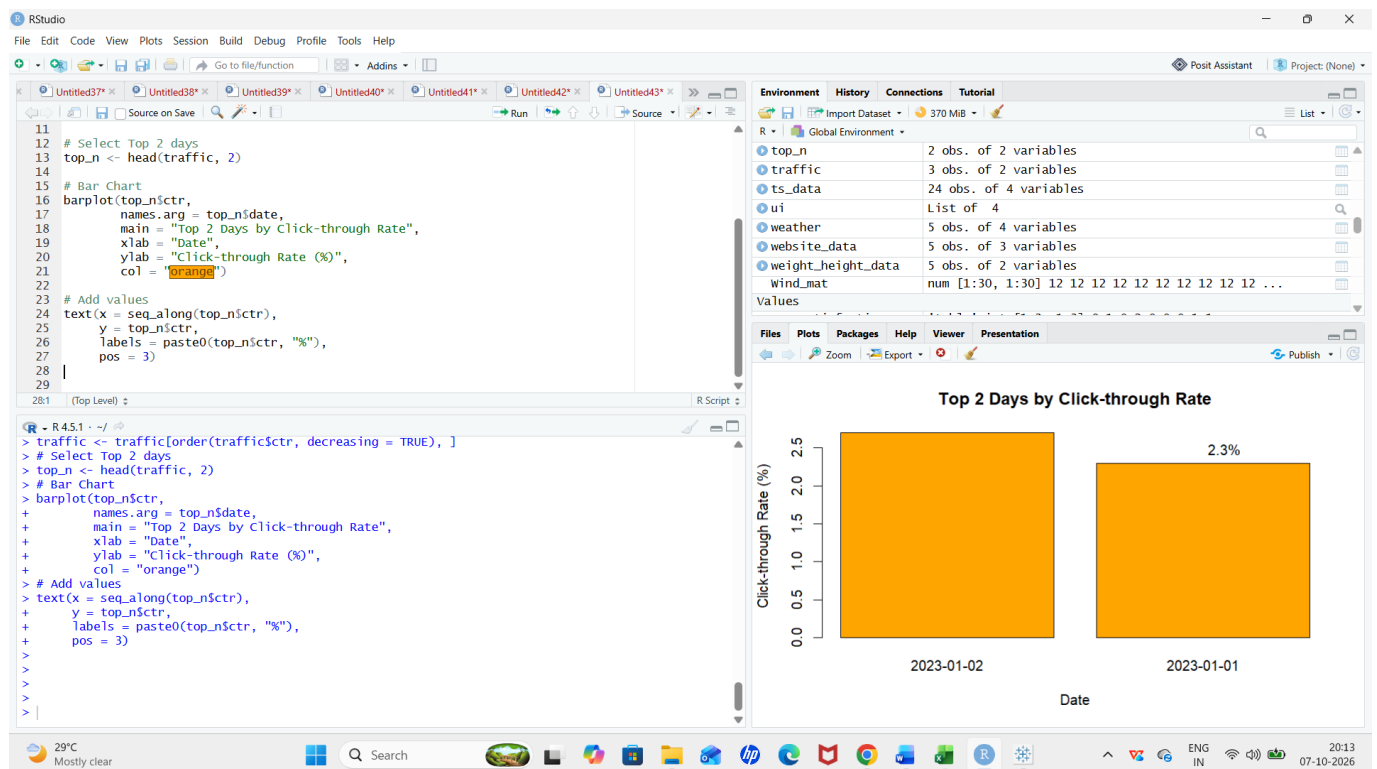
traffic <- data.frame(date, ctr)

traffic <- traffic[order(traffic$ctr, decreasing = TRUE), ]

top_n <- head(traffic, 2)

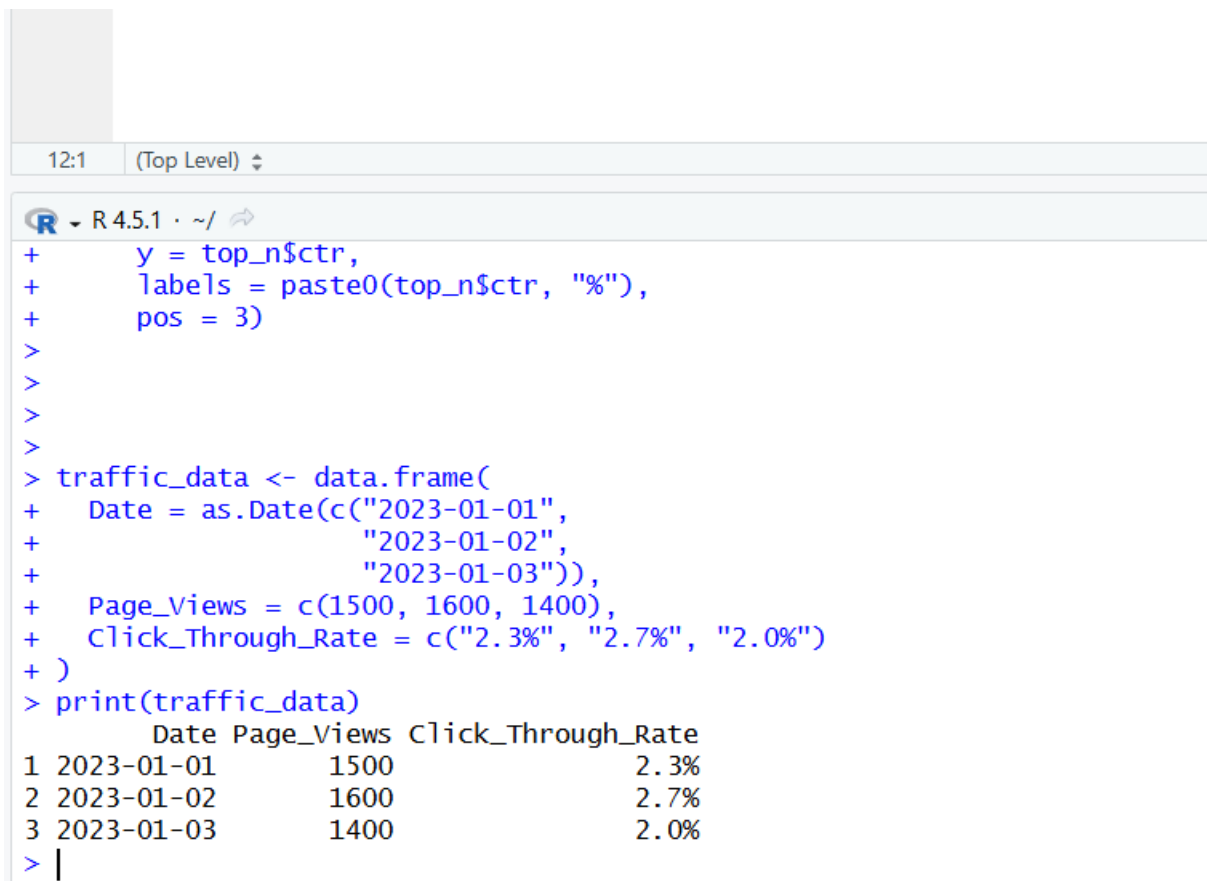
barplot(top_n$ctr,
        names.arg = top_n$date,
        main = "Top 2 Days by Click-through Rate",
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        col = "orange")

text(x = seq_along(top_n$ctr),
     y = top_n$ctr,
     labels = paste0(top_n$ctr, "%"),
     pos = 3)
```



3. Build a table to show the website traffic data for each day. Label the table elements.

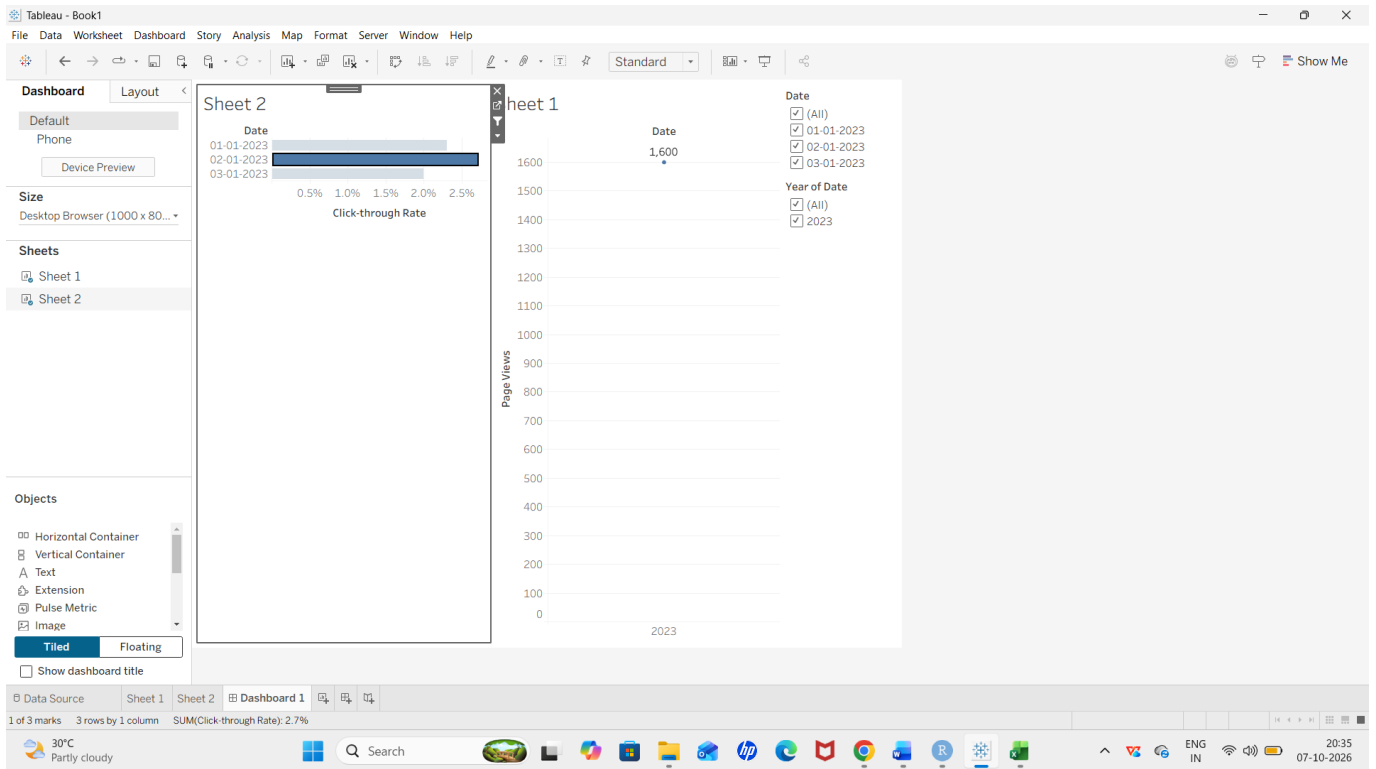
```
traffic_data <- data.frame(  
  Date = as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03")),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
)  
  
print(traffic_data)
```



The screenshot shows an R console window with the following content:

```
12:1 (Top Level) ↕  
R 4.5.1 · ~/ ↗  
+ y = top_n$ctr,  
+ labels = paste0(top_n$ctr, "%"),  
+ pos = 3)  
>  
>  
>  
>  
> traffic_data <- data.frame(  
+   Date = as.Date(c("2023-01-01",  
+                   "2023-01-02",  
+                   "2023-01-03")),  
+   Page_Views = c(1500, 1600, 1400),  
+   Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
+ )  
> print(traffic_data)  
      Date Page_Views Click_Through_Rate  
1 2023-01-01      1500             2.3%  
2 2023-01-02      1600             2.7%  
3 2023-01-03      1400             2.0%  
> |
```

4.



PROCEDURE

1. Enter the website traffic data containing Date, Page Views, and Click-through Rate.
2. Create a line chart to visualize the daily trend in page views.
3. Create a bar chart to display the top days with the highest click-through rates.
4. Create a table to display the website traffic details for each day.
5. Import the dataset into Tableau.
6. Create separate worksheets for the line chart, bar chart, and table.
7. Combine all three worksheets into a Tableau dashboard.
8. Add labels, titles, and filters to make the dashboard interactive.

RESULT

The website traffic data was successfully analyzed using R and Tableau. The line chart shows the daily page-view trend, the bar chart identifies the top days based on click-through rate, and the table displays the complete traffic data. An interactive Tableau dashboard was successfully developed by combining all three visualizations.